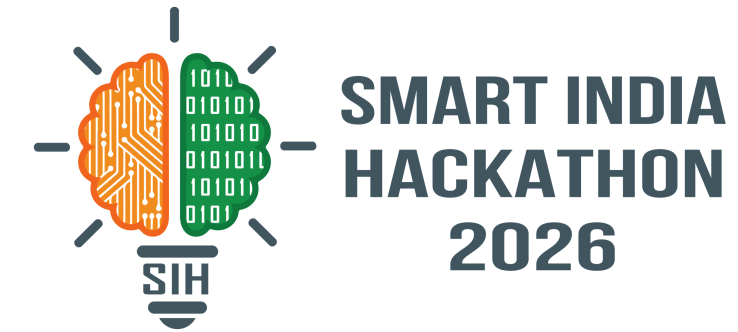


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – 26068**
- **ProblemStatementTitle-WeatherGPT:**
Conservational AI for Weather forecasting, alerts
and Climate Informations
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID- SIH26-AOH-T507**
- **Team Name (Registered on portal)- Quantam**



WEATHERGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Informations

Effects or Problems

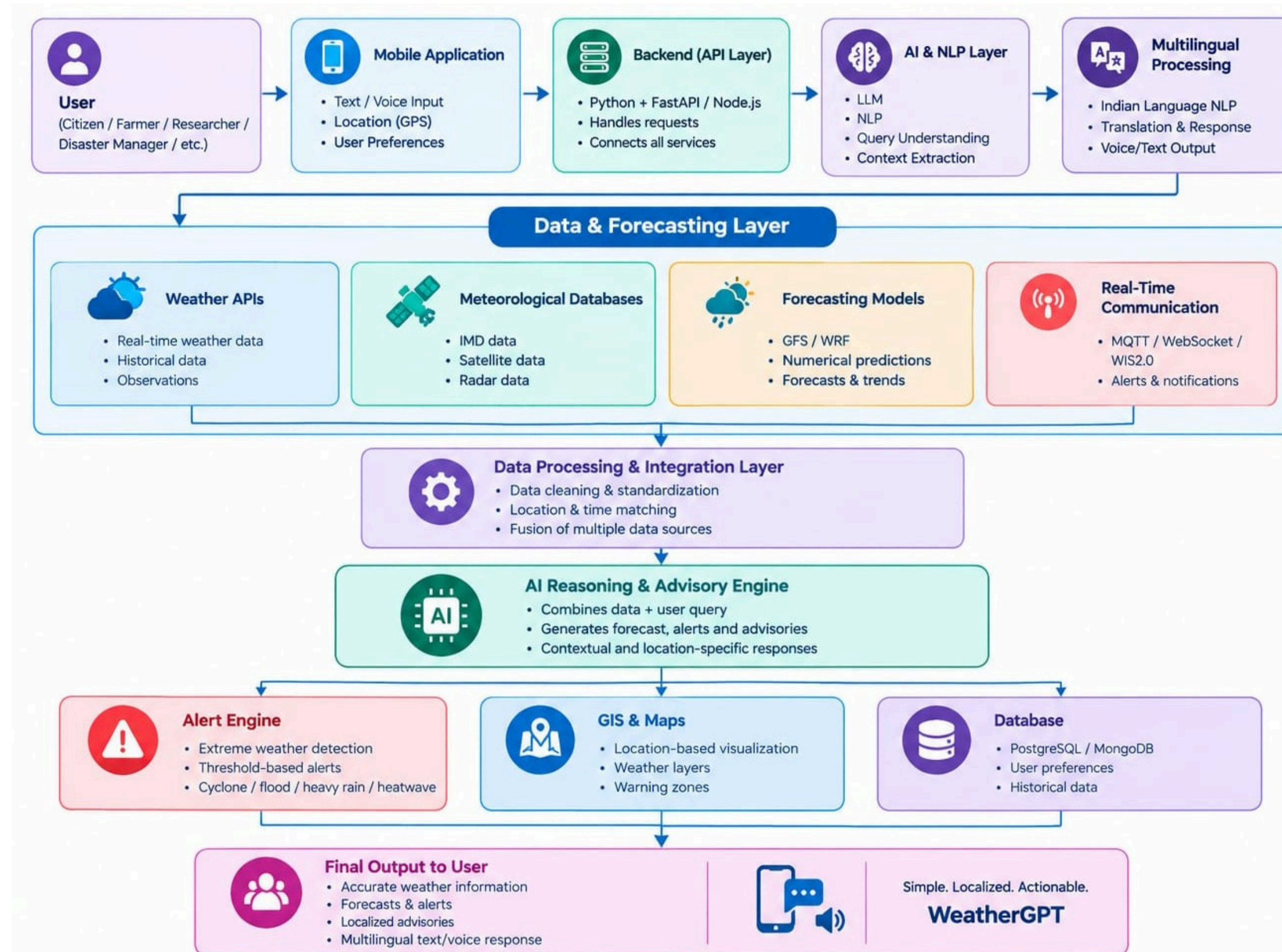
- **Poor coordination**- delay in communication between forecasting agencies and the commoners.
- **Misinformation and rumours**- Unverified informations can spread without any authorization among people.
- **Information Overload**- Lack of abstraction of information that is required.
- **Language Barrier**- Alerts may not be available in their respective languages which reduces the effectiveness.

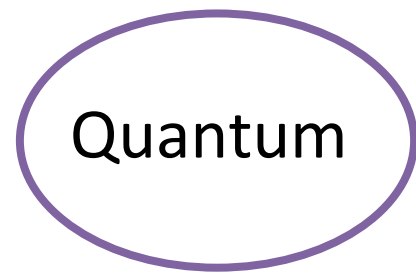
Solution

- **Real time alerts**- Send immediate warnings about the disasters.
- **Location based informations**- Provides alerts and forecasts specific to the users exact area.
- **Verified informations**- Combine data from trusted agencies to reduce misinformation and conflicting updates.
- **Offline/ Emergency support**- Store critical warnings and provide alerts even with limited internet connectivity.

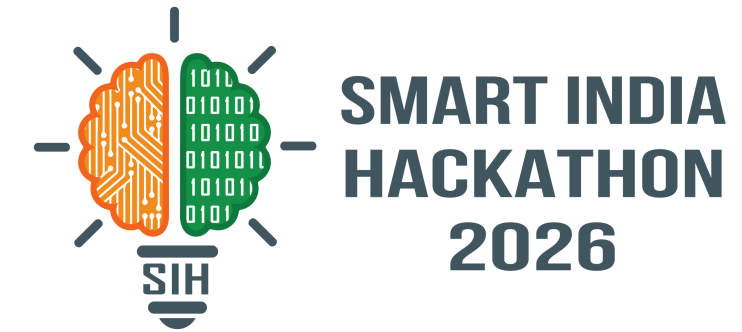
Overall Idea: To create a centralized AI powered location aware communication platform that converts complex weather forecaste data into user friendly data so the people could make faster and safer decision.

TECHNICAL APPROACH





FEASIBILITY AND VIABILITY



THE PROBLEM

Unreliable Weather Information

Weather information from multiple sources can be inconsistent, confusing, or difficult for users to understand.

Delayed Disaster Alerts

Users may not receive timely warnings for extreme weather events such as floods, cyclones, heatwaves and heavy rainfall.

Lack of Location-Specific Data

Generic forecasts may not accurately communicate risks and conditions at the user's exact location.

Language & Accessibility Barriers

Complex weather terminology and limited regional-language support can prevent people from understanding critical warnings.

OUR SOLUTION

AI-Powered WeatherGPT

Combines trusted weather APIs and forecasting models to provide simplified and reliable weather information.

Real-Time Alerts

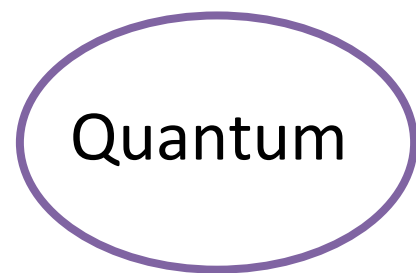
Automatically delivers timely warnings for severe weather and potential disasters.

Location-Based Forecasts

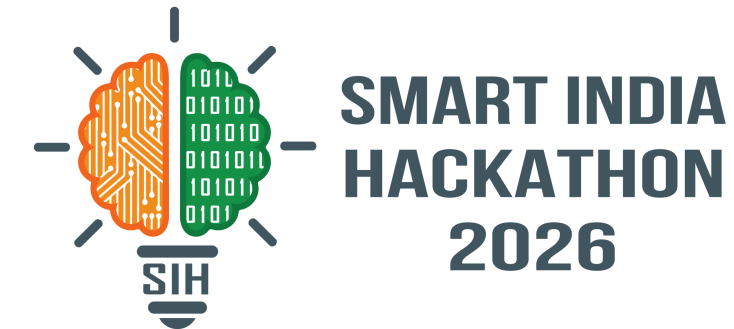
Uses the user's location to provide hyperlocal forecasts, risk levels and relevant safety information.

Multilingual & Accessible

Converts complex weather data into simple, conversational responses with regional-language support.



IMPACT AND BENEFITS



TARGET AUDIENCE IMPACT ANALYSIS

- **FISHERMEN & MARITIME WORKERS**

Get real-time weather and sea-condition alerts.
Plan fishing and travel safely.
Reduce risk from sudden storms and rough seas.

- **FARMERS & RURAL COMMUNITIES**

Receive location-based weather forecasts.
Plan irrigation, sowing and harvesting activities.
Get early warnings for extreme weather.

- **GENERAL PUBLIC**

Get simple, location-specific weather information through conversation.
Receive early warnings for cyclones, floods and heavy rainfall.
Improve local disaster preparedness.
Receive instant alerts about severe weather
Make safer travel and daily-life decisions.

BENEFIT ANALYSIS

- **SAFETY AND ACCESSIBILITY**

Reduces weather-related risks through real-time alerts and early warnings.
Converts complex meteorological data into simple, understandable information.

- **24/7 SUPPORT ACCURACY & TRUST**

AI-based conversational assistance provides weather information anytime, anywhere.
Combines information from verified weather sources to reduce misinformation.

- **INCLUSIVE COMMUNICATION**

Can deliver weather information in user-friendly and regional-language formats, making it accessible to more people.

<https://sachet.ndma.gov.in/>

<https://www.thehindu.com/news/cities/Madurai/tn-alert-app-will-help-in-monsoon-preparedness-say-officials/article68725132.ece>

ScienceDirect.com <https://share.google/3BF8Dsu9povF83sGP>